

**SIKSHA 'O' ANUSANDHAN
(DEEMED TO BE UNIVERSITY)**

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Laboratory Record

**Machine Learning Projects with Python
(CSE 4192)**

Submitted by

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1 Introduction

1.1 Problem Statement

Diabetes mellitus is a progressive chronic metabolic disease marked by persistent hyperglycemia due to insulin deficiency or resistance. Early detection is crucial to prevent microvascular and macrovascular complications. Traditional clinical cutoffs fail to capture non-linear metabolic risk dynamics, while existing machine learning pipelines frequently suffer from data leakage and biological zero-value errors. This project establishes an authoritative, leak-free Multilayer Perceptron (MLP) classification system to accurately predict diabetes onset from patient biometric parameters.

1.2 Aim

To develop, optimize, benchmark, and deploy a data-leakage-free Multilayer Perceptron (MLP) binary classifier for diabetes prediction using the Pima Indians Diabetes Database.

1.3 Objectives

- Conduct rigorous Exploratory Data Analysis (EDA) investigating feature distributions, skewness, and class imbalance (65.1% negative vs. 34.9% positive).
- Implement a leak-free preprocessing pipeline that identifies biological zero artifacts and applies training-fitted median imputation.
- Formulate 16 domain-engineered features (expanding 8 raw attributes into 24 dimensions) capturing WHO BMI bins, ADA glucose stages, and metabolic ratios.
- Benchmark 7 baseline algorithms against an optimized Scikit-Learn Multilayer Perceptron (MLPClassifier).
- Deploy the production model into an interactive Streamlit Cloud web application.

2 Dataset Description

2.1 Dataset Source

The standard Pima Indians Diabetes Database (NIDDK) containing N = 768 female patient records.

2.2 Dataset Features

8 raw physiological inputs: Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, and Age.

2.3 Target Variable

Outcome: Binary label (0 = Non-Diabetic [500 cases, 65.1%], 1 = Diabetic [268 cases, 34.9%]).

2.4 Dataset Statistics

Feature	Type	Mean ± SD	Min	Max	Zeros / Missing (%)
Pregnancies	Discrete	3.85 ± 3.37	0	17	111 (14.45%)*
Glucose	Continuous	120.89 ± 31.97	0 (44)	199	5 (0.65%)
BloodPressure	Continuous	69.11 ± 19.36	0 (24)	122	35 (4.56%)
SkinThickness	Continuous	20.54 ± 15.95	0 (7)	99	227 (29.56%)
Insulin	Continuous	79.80 ± 115.24	0 (14)	846	374 (48.70%)
BMI	Continuous	31.99 ± 7.88	0 (18.2)	67.1	11 (1.43%)
DiabetesPedigree	Continuous	0.47 ± 0.33	0.078	2.42	0 (0.00%)
Age	Discrete	33.24 ± 11.76	21	81	0 (0.00%)

3 Exploratory Data Analysis & 4 Data Preprocessing

3.1–3.8 EDA Insights: Insulin (skewness = 2.27) and DPF (skewness = 1.92) exhibit severe positive skewness. Diabetic individuals show significantly elevated median glucose (140 vs. 107 mg/dL) and BMI (34.3 vs. 30.1 kg/m²).

4.1–4.7 Preprocessing: Zeros in continuous features (Glucose, BP, Skin, Insulin, BMI) were replaced with NaN and imputed using training medians fitted on D_train (N=537): Glucose = 117.0, BP = 72.0, Skin = 29.0, Insulin = 125.0, BMI = 32.0. 16 domain features were engineered (WHO BMI bins, ADA glucose stages, metabolic ratios, interaction terms, log transforms), expanding the feature representation to 24 dimensions, followed by StandardScaler normalization.

5 Methodology & 6 Model Development

5.1–5.4 Architecture: Feed-forward Multilayer Perceptron structured as **Input (24) -> Hidden 1 (64, ReLU) -> Hidden 2 (32, ReLU) -> Output (1, Sigmoid)**.

Trainable Parameters: Layer 1: $24 \times 64 + 64 = 1,600$; Layer 2: $64 \times 32 + 32 = 2,080$; Output: $32 \times 1 + 1 = 33$ -> **Total = 3,713 parameters**.

6.1–6.5 Baselines: Logistic Regression, KNN (k=7), SVM (RBF), Decision Tree, Random Forest (100 trees), Gradient Boosting, and Baseline MLP.

7 Model Evaluation & 8 Model Comparison

Evaluated on the untouched test partition (\$N = 116\$, 76 Negative, 40 Positive cases):

Model	Accuracy	Sensitivity	Specificity	Precision	F1	ROC-AUC	FN	FP
Logistic Regression	79.31%	72.50%	82.89%	0.6905	0.7073	0.8444	11	13
K-Nearest Neighbors	81.90%	77.50%	84.21%	0.7209	0.7470	0.8954	9	12
Support Vector Machine	83.62%	77.50%	86.84%	0.7561	0.7654	0.8808	9	10
Decision Tree	87.93%	82.50%	90.79%	0.8250	0.8250	0.9021	7	7
Random Forest	88.79%	87.50%	89.47%	0.8140	0.8434	0.9431	5	8
Gradient Boosting	89.66%	85.00%	92.11%	0.8500	0.8500	0.9579	6	6
MLP (Baseline)	81.03%	65.00%	89.47%	0.7647	0.7027	0.8625	14	8
MLP (Optimized)	82.76%	75.00%	86.84%	0.7500	0.7500	0.8681	10	10

9 Hyperparameter Tuning & 10–12 Deployment

9.1–9.5 Tuning: Grid Search optimized the MLP: Topology (64, 32), alpha = 0.001, lr = 0.001, batch size = 32 (Validation ROC-AUC = 0.8645).

10–11 Final Evaluation: Retrained on Train+Val (N=652) and tested on untouched Test (N=116), yielding 82.76% Accuracy, 75.00% Recall, 86.84% Specificity.

12.1–12.5 Deployment: Deployed to Streamlit Cloud with an interactive multi-tab interface supporting single-patient triage, counterfactual sensitivity simulation, and batch cohort screening. Verified via automated deployment test suite (7/7 tests passed in 3.2s).

13 Results and Discussion & 14–16 Conclusion

Discussion: The MLP effectively maps complex metabolic interactions into smooth continuous probabilities. In clinical triage, False Negatives present high hazard. Dynamic threshold calibration (tau = 0.35) reduces False Negatives from 10 to 4.

Limitations: Evaluated on Pima Indian female cohort (N=768); multi-ethnic external validation is warranted.

Future Scope: Tabular Transformers (TabNet), explainable AI (SHAP), and edge mobile deployment.

References

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Appendix D: Contributions of Group Members

Sl.	Group Member	Regd. No.	Role / Responsibility	Contribution (%)
1	Tribhuwan Singh	2341019538	ML Lead, MLP Architecture & Pipeline Architect	25%
2	Surajit Sahoo	2341019165	Data Preprocessing, Imputation & Baseline Modeling	25%
3	Anwesha Srichandan	2341019594	EDA, Feature Engineering & Hyperparameter Search	25%
4	Priti Rani Maity	2341013065	Model Evaluation, Web Deployment & Report Preparation	25%